⚙️ METALS AND NON-METALS

TOPPER NOTES

🧪 Chemistry Chapter 3 | 🛡️ Reactivity Masterclass | 🌻 Hinglish Edition

1. Physical Properties & Differences ⚖️

Elements ko unke physical aur chemical behavior ke base par Metals aur Non-metals mein divide kiya jata hai.

Feature	Metals (Dhaatu) 🪙	Non-Metals (Adhaatu) ⚫
Physical State	Solid at room temp (Except Mercury, which is liquid).	Solids, liquids (Bromine), or gases.
Lustre & Malleability	Shiny (lustrous) and malleable (can be beaten into sheets).	Non-lustrous (dull) and brittle (break easily).
Ductility	Ductile (can be drawn into wires). E.g., Gold is most ductile.	Non-ductile.
Conduction	Good conductors of heat and electricity (Silver & Copper best).	Bad conductors of heat/electricity (Except Graphite).

⭐ **Exceptions to Remember:** Sodium (Na) and Potassium (K) soft metals hain jinhe knife se kata ja sakta hai! Mercury ek aisi metal hai jo room temperature par liquid hoti hai.

2. Chemical Properties of Metals ⚡

A. Reaction with Oxygen:

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Metal + Oxygen $\rightarrow$ Metal Oxide (Basic in nature) Example: $2Mg + O_2 \rightarrow 2MgO$ (Magnesium oxide - basic) *Note:* Kuch metal oxides jaise $Al_2O_3$ aur $ZnO$ **Amphoteric** hote hain (ye acid aur base dono ke sath react karte hain).
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B. Reaction with Water:

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Metal + Water $\rightarrow$ Metal Oxide / Hydroxide + Hydrogen gas ($H_2$) - Potassium & Sodium react violently with cold water. - Calcium reacts less violently. - Iron and Aluminum react only with steam.
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C. Reaction with Acids:

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Metal + Dilute Acid $\rightarrow$ Salt + Hydrogen gas ($H_2$) *Exception:* Nitric acid ($HNO_3$) hydrogen gas release nahi karta kyunki ye ek strong oxidising agent hai jo $H_2$ ko $H_2O$ mein convert kar deta hai (Magnesium and Manganese are exceptions jo very dilute $HNO_3$ ke sath $H_2$ dete hain).
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3. The Reactivity Series (Sakriyata Shreni) 📊

Metals ko unki reactivity ke decreasing order mein arrange karne par **Reactivity Series** banti hai.

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K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Au (Most Reactive $\rightarrow$ Least Reactive)
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> 💡 **Short Trick to Remember Reactivity Series:**
> **K**alu **Na**ne **Ca**ar **Mg** **Al**se **Zn**e **Fe**eki **Pb**ar **[H]**igh **Cu** **Hg** **Ag** **Au**
> *(K, Na, Ca, Mg, Al, Zn, Fe, Pb, H, Cu, Hg, Ag, Au)*

4. Ionic Compounds (Electrovalent Compounds) ⚛️

Transfer of electrons from a metal to a non-metal forms ionic compounds (e.g., $NaCl$).

- **Properties:** High melting and boiling points, soluble in water, conduct electricity in molten or aqueous state.

5. Metallurgy & Extraction of Metals ⛏️

Ores (khaadan se milne wale minerals jisme metal zyada amount mein ho) se pure metal nikalne ki process ko **Metallurgy** kehte hain.

Steps in Extraction:

- **1. Enrichment of Ore (Concentration):** Gangue (impurities) ko remove karna.
- **2. Extracting Metals from Concentrated Ore:**
  - **Low Reactivity:** Heating sulphide ores directly (Roasting). E.g., Cinnabar (HgS).
  - **Medium Reactivity:** Calcination (absence of O_2 for carbonate ores) & Roasting (presence of O_2 for sulphide ores), followed by reduction with carbon.
  - **High Reactivity:** Electrolytic reduction (e.g., $NaCl$ molten state mein electrolysis).
- **3. Refining of Metals:** Electrolytic refining (Pure metal cathode par, impure metal anode par).

6. 🌟 Dedicated Formulas & Important Reactions Corner

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1. Amphoteric Oxides: Al_2O_3, ZnO 2. Roasting (Sulphide Ore): 2ZnS + 3O_2 $\xrightarrow{\Delta}$ 2ZnO + 2SO_2 3. Calcination (Carbonate): ZnCO_3 $\xrightarrow{\Delta}$ ZnO + CO_2 4. Reduction by Carbon: ZnO + C $\rightarrow$ Zn + CO 5. Thermite Reaction: 3Fe_2O_3 + 2Al $\rightarrow$ 2Fe + Al_2O_3 + Heat (Molten Iron used for railway tracks) 6. Rusting Formula: Fe_2O_3 \cdot xH_2O (Hydrated iron(III) oxide)
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7. Corrosion & Prevention 🛡️

Metals par moisture aur oxygen ki wajah se layer jam jana corrosion kehlata hai (Iron par rust, Copper par green coating, Silver par black coating). Prevention: Painting, galvanizing, anodising, alloying.

8. ⚡ Quick Revision Checklist (Last-Minute Summary) ✅

- [ ] **Liquid Metal:** Mercury (Hg) | **Liquid Non-Metal:** Bromine (Br_2).
- [ ] **Amphoteric Oxides:** Al_2O_3 and ZnO (React with both acid & base).
- [ ] **Roasting:** Done for sulphide ores in presence of excess air.
- [ ] **Calcination:** Done for carbonate ores in limited air.
- [ ] **Galvanization:** Coating iron with a thin layer of Zinc to prevent rusting.

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